Containerization-and-DevOps

🐳 Docker API & Image Management – Class Task

🛠️ Technologies Used

- Docker
- WSL (Windows Subsystem for Linux)
- cURL
- Nginx

⚙️ Steps Performed

1️⃣ Start WSL

```
wsl
```

2️⃣ Verify Docker Socket

```
ls -l /var/run/docker.sock
```

3️⃣ Check Docker Daemon (Ping Test)

```
curl --unix-socket /var/run/docker.sock http://localhost/_ping
```

Output:

```
OK
```

4️⃣ Get Docker Version using API

```
curl --unix-socket /var/run/docker.sock http://localhost/version
```

5️⃣ List Running Containers using API

```
curl --unix-socket /var/run/docker.sock \
http://localhost/v1.44/containers/json
```

Output:

```
[]
```

6️⃣ Check Running Containers (CLI)

```
docker ps
```

7️⃣ Pull Nginx Image

```
docker pull nginx
```

⚠️ Notes

- Docker socket is used for direct communication with Docker daemon
- cURL allows accessing Docker REST API without Docker CLI
- No containers were running initially
- Nginx image was successfully downloaded from Docker Hub

✅ Outcome

- Verified Docker daemon using socket
- Accessed Docker API endpoints

- Retrieved system and container information
- Successfully pulled Nginx image

📸 Screenshots

> all screenshots below:

```
PS C:\Users\srist> wsl
sristee@Sristee:/mnt/c/Users/srist$ ls -l /var/run/docker.sock
srw-rw---- 1 root docker 0 Feb  3 10:07 /var/run/docker.sock
sristee@Sristee:/mnt/c/Users/srist$ curl --unix-socket /var/run/docker.sock http://localhost/_ping
OKsristee@Sristee:/mnt/c/Users/srist$
```

```
PS C:\Users\srist> wsl
sristee@Sristee:/mnt/c/Users/srist$ ls -l /var/run/docker.sock
srw-rw---- 1 root docker 0 Feb  3 10:07 /var/run/docker.sock
sristee@Sristee:/mnt/c/Users/srist$ curl --unix-socket /var/run/docker.sock http://localhost/_ping
OKsristee@Sristee:/mnt/c/Users/sristcurl --unix-socket /var/run/docker.sock http://localhost/versionon
{"Platform":{"Name":""},"Components":[{"Name":"Engine","Version":"28.2.2","Details":{"ApiVersion":"1.50","Arch":"amd64",
```

```
sristee@Sristee:/mnt/c/Users/srist$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
sristee@Sristee:/mnt/c/Users/srist$ curl --unix-socket /var/run/docker.sock \
  http://localhost/v1.44/containers/json
[]
sristee@Sristee:/mnt/c/Users/srist$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0c8d55a45c0d: Pulling fs layer
173e7a5d3717: Pulling fs layer
2711d25abbb0: Pulling fs layer
359cde133485: Waiting
acc398fdf80e: Waiting
bc4d011570c3: Waiting
1eb69ddd819c: Waiting
```

```
sristee@Sristee:/mnt/c/Users/srist$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0c8d55a45c0d: Downloading [====>                                          ]  2.482MB/29.78MB
173e7a5d3717: Downloading [=>                                             ]  1.028MB/33.07MB
2711d25abbb0: Download complete
359cde133485: Download complete
acc398fdf80e: Download complete
bc4d011570c3: Waiting
1eb69ddd819c: Waiting
```